EXERCICE 1

Let x be real and $g(x) = x^2 + 2x + 3$.

1. a is any given real number. Express the rate of change of g at a as a function of a and h.
2. Deduce the expression of $g'(a)$ as a function of a.
3. Use the previous result to find the value of a for which the tangent line of the curve of g at point $\Big(a, g(a)\Big)$ has slope 3. Give an equation of this line.
4. Does the curve of g have a horizontal tangent somewhere? If so, for what value of a?